



Review

Neurobiological and neuroimaging biomarkers: A narrative review of precision medicine for diagnosing neurodegenerative disorders

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ABSTRACT

Markers provide information about brain function, structure, and activity and are often used to study neurological and psychiatric conditions. This is crucial for diseases such as Parkinson's disease, Alzheimer's disease, autism, and Huntington's disease. Neuromarkers describe distinct biological markers of these states, allowing for a new standard of earlier and more specific diagnosis than traditional symptom-based approaches. Herein, this review summarizes the current state of research on neurobiological and neuroimaging biomarkers in neurodegenerative diseases. Techniques, such as magnetic resonance imaging, positron emission tomography, and functional magnetic resonance imaging, play a crucial role in characterizing the structural, functional, and molecular imaging changes in the brain, offering new potential approaches for developing more precise treatment strategies tailored to individual patients. Neuroinflammation has become one of the most studied common pathological mechanisms involved in those diseases. Neurobiomarkers have been shown to reflect underlying immune phenomena. The presence of these neuroimaging markers, such as microglial activation or cytokine levels, can reveal a patient's inflammatory status, allowing the differential diagnosis of disease states and tracking disease-specific pathophysiology. In Alzheimer's and Parkinson's diseases, for example, imaging biomarkers of tau and alpha-synuclein help differentiate subtypes and predict the potential utility of treatments, while in autism and Huntington's disease, systems-level functional connectivity patterns serve as clues to underlying neural dysfunction. The clinical application of neuromarkers holds the potential to promote precision medicine, with the opportunity to develop treatments specifically for the neuromodulatory substrates of various diseases. Nevertheless, challenges remain in standardization and predictive validation of biomarkers between different race groups. Future studies should develop multimodal imaging, artificial intelligence-based analysis, and large-scale longitudinal studies to move the innovations toward practical diagnosis and therapeutic approaches, thereby improving the prognosis of patients with either neurodegenerative or neurodevelopmental disorders.

1. Introduction

Neuroscience markers, also called neuromarkers, are quantifiable parameters that give insights into the structure, functioning, and biochemical activity of the human brain.^{1,2} Neuromarkers are central in elucidating the molecular and physiological underpinnings of neurological and psychiatric disorders, such as Alzheimer's disease, Parkinson's disease, autism spectrum disorder, and Huntington's disease.^{3,4} Importantly, in neuroscience studies, neuromarkers are used as

diagnostic and prognostic aids, filling the void between molecular neuropathology and the clinicopathological disease continuum.^{5–7} Neuromarkers offer objective and quantifiable data as opposed to conventional symptoms-based rating scales, making it possible to offer earlier diagnosis, patient stratification, and to monitor disease progression.⁸

Neuromarkers have also been identified due to the recent breakthroughs in neuroimaging and molecular biology.^{9–11} Multiple techniques, including magnetic resonance imaging (MRI), functional MRI

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(fMRI) and positron emission tomography (PET), have enabled the characterisation and quantification of structural, metabolic and functional changes associated with neural disorders.^{12–14} Hallmark biomarkers of neurodegeneration and neurodevelopmental dysfunction include imaging of tau, amyloid- β facilitating the diagnosis and differentiation of Alzheimer's disease subtypes; imaging of α -synuclein contributes to the detection of early Parkinson's disease pathology.¹⁵ Congruently, neuroinflammatory markers, such as cytokines, chemokines or microglial activation, found an increasing number of neuromarkers being implicated as central to neurodegeneration or neurodevelopmental dysfunction.^{16,17}

Recently, complementary novel research has highlighted the potential benefits of multimodal imaging, artificial intelligence (AI) algorithms and analytics and large-scale longitudinal databases.^{18,19} As a result, it is projected that the specificity, reproducibility and clinical utility of neuroscience markers will increase. Ultimately, the development of neuroscience markers will support precision medicine, i.e., disease biomarkers that are adapted to patient specific diagnostic and treatment approaches targeted to the disease specific neuromodulatory and molecular pathogenic substrates.²⁰ The purpose of this review is to comprehensively examine the current understanding of neuroscience markers (neuromarkers) in the context of neurodegenerative and neurodevelopmental diseases, emphasizing their diagnostic, prognostic, and therapeutic potential. This paper aims to integrate evidence from neuroimaging, molecular, and inflammatory biomarkers to highlight their role in advancing precision medicine. The central hypothesis of this review is that multi-dimensional neuromarkers combining structural, functional, and molecular indicators can provide a more accurate representation of brain pathology, thereby improving early diagnosis, patient stratification, and individualized treatment strategies across neurological disorders (Fig. 1).

2. Search strategy

We conducted an extensive literature search to identify articles reporting on neuroscience markers (neuromarkers), neuroinflammation, and neuroimaging biomarkers in neurodegenerative/developmental disorders. The search was conducted in PubMed, Scopus, Web of Science, and Google Scholar databases for the period 2020–2025.

3. Mechanism of neuromarkers and neuroimaging in precision medicine for neurodegenerative disorders

Neuromarkers and neuroimaging together outline a mechanism process that can be used to diagnose and monitor neurodegenerative and neurodevelopmental diseases. The process starts with disease-specific pathological molecular and cellular alterations, including misfolded protein aggregation (e.g., tau in Alzheimer's disease or α -synuclein in Parkinson's disease), loss of neurons, synaptic disorganization, and neuro-inflammation.^{21,22} These pathological changes will provoke immune responses with microglial activation and changing cytokine profile, which can be reflected by certain neurobio-markers in the cerebrospinal fluid (CSF), blood, or molecular imaging.²³

Neuroimaging methods render these molecular and cellular changes into visible structural, functional, and molecular expression. Brain atrophy and white matter integrity can be visualized with MRI, connectivity and network dynamics are revealed by using fMRI, whereas protein aggregates and neuroinflammatory processes can be imaged by PET.^{23,24} Integrating these neuromarkers with imaging allows clinicians to detect disease earlier, in different forms, and as it progresses. The combination of these multimodal methods, underpinned by AI and machine learning, further improves the predictive capabilities of systems to facilitate more precise interventions for individual patients.^{21,25} This mechanistic pathway explains the molecular pathology, immune signaling, and imaging instrumentation that coalesce to provide a precision-based method in neurodegenerative diseases (Fig. 2).

4. Neuromarkers measure brain function and dysfunction

Biomarkers are measurable biological characteristics that may hold promise for the study of brain physiology, health, and disease. Such markers could be (neuro-) physiological, biochemical, genetic, or -neuroimaging measures that contribute to the characterization of neuropsychiatric or neurological diseases.^{5,26,27} For example, fMRI and electroencephalography (EEG) are able to detect pathological activation of the brain associated with disorders like depression, schizophrenia, and Alzheimer's disease.^{28–30} Since they can serve as an objective tool for diagnosis, neuromarkers are a means of increasing the accuracy of mental health assessments and reducing the reliance on self-reports of symptoms.^{31,32} One of the exciting applications for neuromarkers is the early stages of disease. Regional alterations in brain chemistry and connectivity are common in neurodegenerative disorders such as Parkinson's disease and Alzheimer's disease long before any clinical symptoms are apparent.^{33,34} Markers such as β -amyloid plaques, tau proteins, and dopamine transporter can be symptomatic of future disease development, which opens the door to early treatments.^{35,36} In addition, machine learning algorithms have a growing presence in neuroimaging analysis, and accordingly may also, in part, retard the error rate in neuromarkers predictive ability as well as for conditions such as epilepsy and traumatic brain injury.^{37,38} Neuromarkers are essential for personalized medicine and therapeutic decision-making.³⁹ For instance, EEG markers can predict antidepressant responses in major depressive disorder.^{40,41} Similarly, neuroimaging could aid in identifying patients who would be more likely to benefit from cognitive-behavioral therapy *versus* pharmacotherapy when the response is compared.^{42,43} Great for the patient and great for the prescribing process, too, which means less trial-and-error prescribing that leads to side effects and poor success.

Furthermore, neurofeedback treatments are already examining neuromarkers that, when their cerebral activity is in good condition in real time, the patients may self-regulate its dysfunction.^{44–46} Despite the potential, neuromarkers must overcome several obstacles. Inter-individual variability in brain anatomy and function may also affect the detection of biomarkers.^{47,48}

Moreover, the expensive, inaccessible advanced neuroimaging techniques are limitations for the general use of such methods in clinical practice. There are also ethical issues, including privacy concerns around neural data and the possible misuse of predictive biomarkers in the context of employment or insurance. Novel technologies add to the range of neuromarkers that can be discovered.^{49,50} Wearable EEG devices, portable near-infrared spectroscopy, and liquid biopsy devices for blood-based detection of brain-derived biomarkers are making neuromarker evaluation more available.⁵¹ AI is also advancing the processing of the massive neurobiological dataset and revealing new biomarkers for diseases such as autism and post-traumatic stress disorder.^{52,53} With the maturation of these technologies, however, this may change significantly as neuromarkers may become the foundation not only of diagnosis and treatment but also of preventive healthcare systems aimed at identifying individuals at risk before the onset of a condition or disease.^{54,55} Neuromarkers are considered a revolutionary innovation in neuroscience because they are an objective, rigorous measure of the health of brain.⁵ Their uses range from early diagnosis of diseases to personalized treatment plans and are becoming increasingly numerous.⁵⁶ However, technical, ethical, and accessibility issues will need to be resolved before they become part of mainstream medicine. Interdisciplinary cooperation between neuroscientists, clinicians, and engineers should further the field, leading to the realization of the full potential of neuromarkers for neurocognitive disorders.

5. Role of neuromarkers in early diagnosis

Neuromarkers, or biomarkers developed from neurological results, are essential in the early identification of neurological and psychiatric

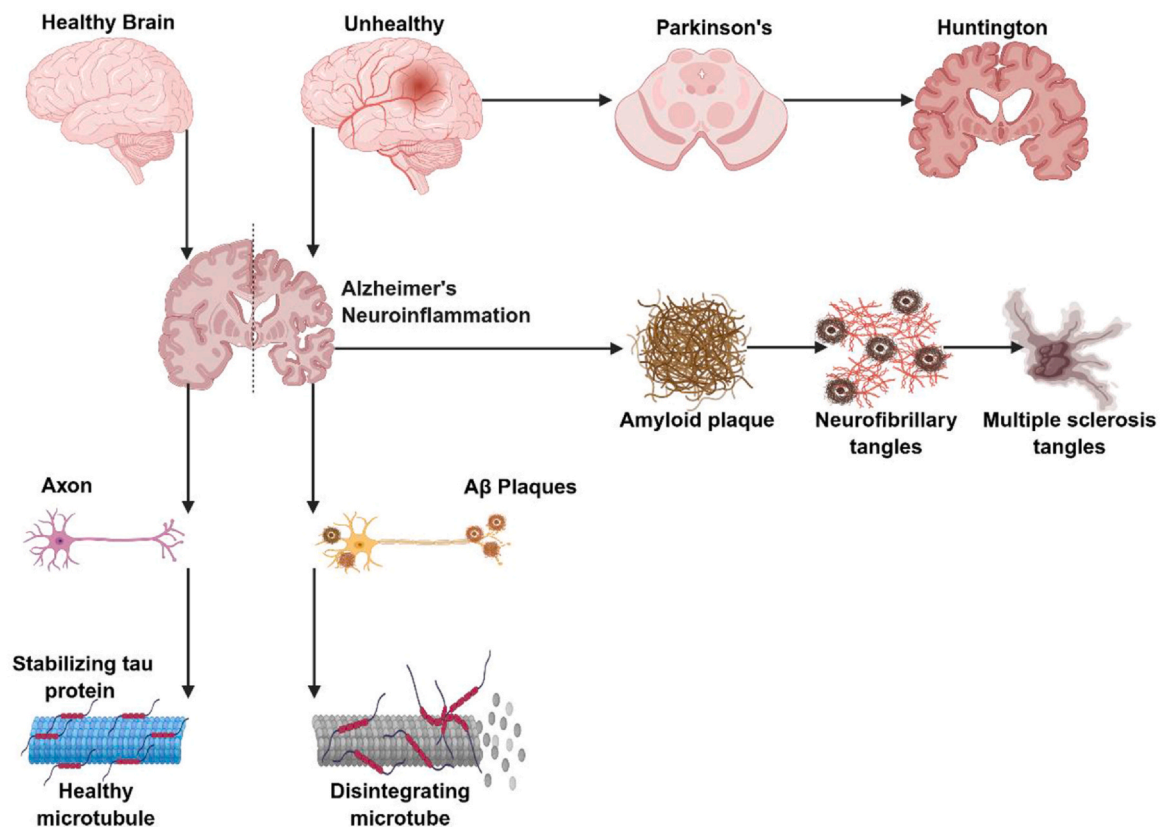


Fig. 1. Pathologies of neuroinflammation, Alzheimer's, Parkinson's, and Huntington's diseases. A β : Amyloid-beta.

The progression from normal brain to various pathological states associated with major neurodegenerative diseases. Healthy condition of the brain; strong microtubules and proper functioning tau proteins allow for neuronal integrity and synaptic activity to be maintained. By contrast, neuroinflammation emerges as a pivotal disease mechanism underlying different types of neurodegeneration. In Alzheimer's disease, persistent neuroinflammation is associated with the build-up of amyloid-beta (A β) plaques and tau hyperphosphorylation that gives rise to the formation of neurofibrillary tangles and axonal degeneration. These changes at the molecular level lead to the collapse of microtubular stability and synaptic failure that mediate cognitive disabilities. Analogous neuroinflammatory cascades occur in Parkinson's and Huntington's diseases, where misfolded proteins and loss of dopaminergic or striatal neurons propagate the process of neurodegeneration. Also, multiple sclerosis presents inflammatory-mediated demyelination and axonal loss, which highlight the role of neuroinflammation as a common pathologic mechanism among diseases. Together, the figure shows the interrelated protein aggregation, microtubule destabilization, and immune activation that cause neuronal dysfunction and progressive brain atrophy seen in the neurodegenerative diseases. Created using BioRender.com.

diseases.^{57,58} These are disease risk or onset markers beyond clinical symptoms through genetic, biochemical, neuroimaging, or electrophysiological measures that allow the detection of early signs of disease risk or onset.^{59,60} Indeed, amyloid-beta (A β) and tau CSF proteins are established neuromarkers of Alzheimer's disease, which may be detected years before the onset of cognitive symptoms.^{61–63} An early diagnosis based on neuromarkers paves the way for early interventions, which could slow down disease progression and promote better patient outcomes.^{54,64} Neuroimaging techniques, including fMRI and PET, have enabled means to detect early brain changes seen in conditions ranging from Parkinson's disease to schizophrenia.⁶⁵ EEG-derived neuromarkers, including atypical groups of theta-gamma coupling, have been proposed in individuals at risk for psychosis.^{66,67} As well these non-invasive surrogate markers offer gross potential in understanding neural dysfunction that aid in early and accurate diagnosis.⁶⁸ Inclusion of machine learning for the neuromarker analysis further increases the overall prediction accuracy for personalized medicine.^{69,70} However, the application of neuromarkers in early diagnosis has been confronted with a number of bottlenecks.^{11,71} Variability in the expression of biomarkers and the expense of advanced neuroimaging limit access.^{72,73} In addition, ethical issues such as the psychological impact of early risk disclosure should be addressed.⁷⁴ With these neuromarkers, doctors could shift from being reactive to preventive, even improving the quality of life for these high-risk groups.

6. Neuroimaging: techniques and applications

Neuroimaging encompasses various techniques used to visualize the structure and function of the living brain.^{75,76} These methodologies are pivotal in studies related to neurologic disorders, cognitive functions, and cerebral development.^{77,78} These methodologies are pivotal in studies related to neurologic disorders, cognitive functions, and cerebral development.^{79,80} Neuroimaging has revolutionized neuroscience by enabling a non-invasive look at the brain and being useful for diagnosis, treatment planning, and research.^{81,82} Structural neuroimaging techniques (e.g., MRI and computed tomography scans) are crucial for the evaluation of brain pathology, namely that of tumors, infarctions, or traumatic brain injuries.⁸³ MRI for soft tissue contrast, a high-resolution imaging technique relying on magnetic fields and radio waves, is particularly adapted for soft tissue contrast.^{84–86} Computed tomography scans take less time and are more common, so they are frequently used in emergency situations.⁸⁷ These are useful in enabling clinicians to evaluate brain atrophy, lesions, and vascular anomalies. In addition, development of diffusion tensor imaging has enabled the mapping of white matter tracts and gains in insight of neural connections.^{88,89} Functional neuroimaging methods quantify brain function by changes in blood flow, metabolism, or electric activity.^{90,91} When a patient is completing an MRI, fMRI maps the flow of oxygenated blood, highlighting the areas of the brain at work when the person is conducting tasks or at rest.^{92,93} PET scan uses radioactive tracers to track metabolic activity, which can

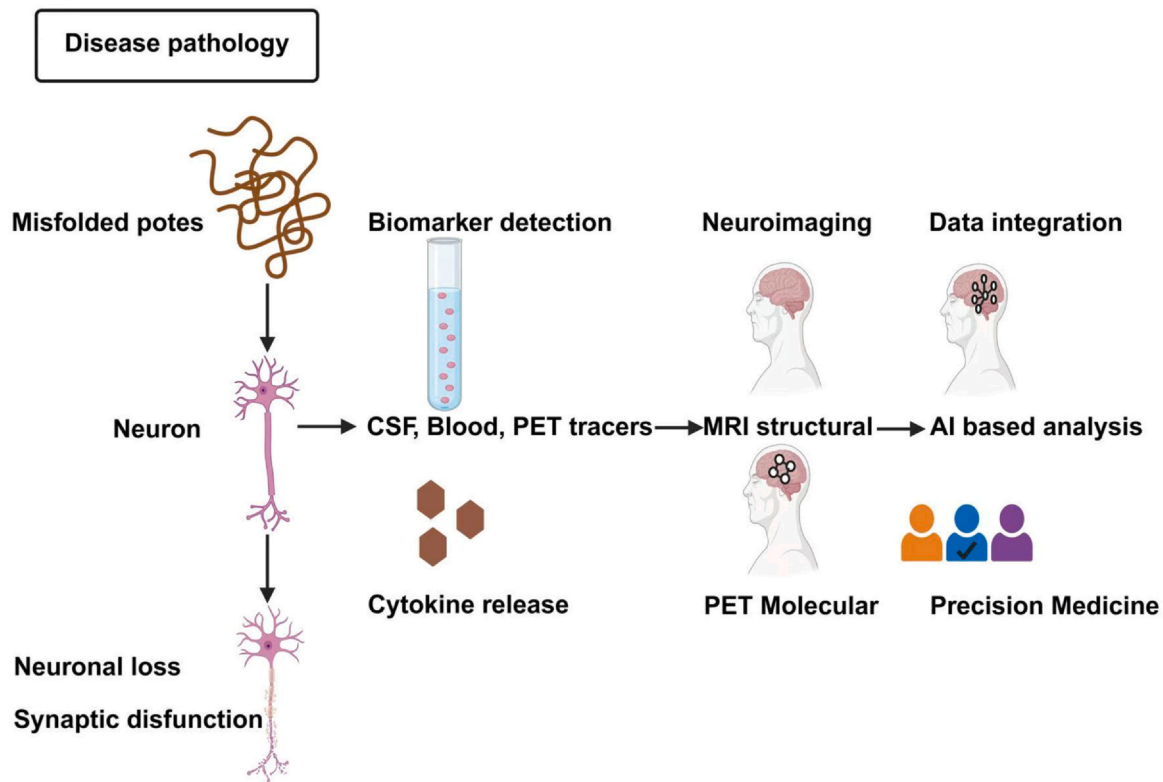


Fig. 2. Mechanism of neuromarkers and neuroimaging. CSF: Cerebrospinal fluid; MRI: magnetic resonance imaging; PET: positron emission tomography. This schematic depicts the mechanistic route from molecular pathology to precision medicine applications. Starting with the pathologic anatomy of disease, aberrantly folded proteins (e.g., tau or α -synuclein) trigger neurodegeneration by causing neuronal loss and impaired synaptic function. These pathologic alterations induce neuroinflammatory changes like cytokine release or microglial activation that may be observed as biomarkers, either in the cerebrospinal fluid (CSF), blood, or with positron emission tomography (PET) tracers. Neuroimaging techniques, such as magnetic resonance imaging (MRI) for structural examination and PET for molecular imaging, image disease-specific brain changes. Artificial intelligence (AI)-analysis combining multimodal imaging and biomarker data will bring added value in disease subtypes definition, progression prediction, and personalized precision medicine approach, leading to individual diagnosis/treatment strategies for neurodegenerative diseases. Created using BioRender.com.

aid in studying neurodegenerative diseases, such as Alzheimer's disease.^{94,95} EEG measures electrical activity with high temporal resolution and is used in the diagnosis of epilepsy and for cognitive tasks.^{96,97} These data-oriented new tools contribute to a better knowledge of brain networks and disorders.^{98,99} In medicine, neuroimaging is of fundamental importance for the diagnosis and monitoring of neurological and psychiatric diseases.^{95,100} Psychiatry, neuroimaging uncovers biomarkers for depression and schizophrenia and supports personalized medicine.^{101,102} In addition, treatment response is followed by neuroimaging, such as in oncology, to track tumor growth.^{10,103} These are just a few examples that illustrate the place within the current practice of medicine and the role that neuroimaging has played in cognitive neuroscience in advancing our understanding of brain structure-function relationships.^{104–106} Empirical evidence using fMRI has shown neuro-correlates of memory, attention, and decision-making.^{107,108} Resting-state fMRI concentrates on intrinsic brain networks and has been applied to examine autism and attention deficit hyperactivity disorder as well.^{109,110} Finally, neuroimaging informs the field of brain-computer interface research and the possibility of communication for patients with paralysis.¹¹¹ The present findings demonstrate how neuroimaging can aid in unraveling the mysteries of the human brain.^{2,112} Innovative methods such as ultra-high-field MRI and AI, are advancing accuracy and analysis of neuroimaging.^{113,114} Ethical concerns have been raised about the privacy of data, the opportunity for incidental findings, and the potential for misuse in lie detection or neuromarketing.^{115,116} Tackling these challenges will require interdisciplinary cooperation and regulatory frameworks. As brain imaging technology advances, it opens new frontiers, but also requires careful use to ensure it serves both smart science and a smarter society.

7. Role of precision medicine in neurodegenerative diseases

It is a different practice of medicine called personalized medicine, catered to your specific genetics and personalized according to your environment and lifestyle.^{117,118} This approach is desirable in the context of neurodegenerative pathologies, such as Alzheimer's disease, Parkinson's disease, and, remarkably, amyotrophic lateral sclerosis, which are complex and highly heterogeneous.^{119,120} With the aid of next-generation genomic sequencing, biomarker screening, and machine learning, precision medicine aims to distinguish multiple disease subtypes, predicting the course of events and finding therapies based on the origin of the disease.^{121–123} This breaking away from a one-size-fits-all philosophy when it comes to precision treatments has the potential to fundamentally change the neurodegenerative fields as we know them.^{124–126} The identification of risk genes is a significant medical application for precision medicine in (neurodegenerative) diseases.

Additionally, it has been linked to mutations with disease risk and progression (e.g., Apolipoprotein E in Alzheimer's disease, LRRK2 in Parkinson's disease, C9ORF72 in amyotrophic lateral sclerosis).^{127,128} Genetic profiling allows for early risk assessment and may also help develop gene-targeted therapies, such as antisense oligonucleotides targeting against SOD1 mutation in SOD1 familial amyotrophic lateral sclerosis.¹²⁹ In addition, the discovery of polygenic risk scores to predict disease predisposition and to guide early intervention of preventive care emphasizes the significance of early-phase precision medicine.^{130,131} The discovery of biomarkers is also a core part of precision medicine in neurodegenerative disorders. Advances in neuroimaging (e.g., A β on PET imaging) and CSF biomarkers (e.g., tau and neurofilament light chain) offer the prospect of an earlier and more

accurate diagnosis, even when patients are asymptomatic.¹³² Such biomarkers improve diagnostic accuracy and can also track drug response in clinical trials.^{133,134} For instance, by testing alpha-synuclein seed amplification assays in Parkinson's disease, it has been demonstrated that pathologic changes can be detected many years before clinical onset as distinct from post-mortem examination or antemortem neuroimaging that only detect the manifested pathology,¹³⁵ thus providing the evidence that a preventive therapy may really become feasible. While this approach might still have promising prospects in the identification of new drugs for neurodegenerative diseases, there are several hurdles to overcome, such as genetic heterogeneity and lack of a definitive cure. The majority of patients have no known genetic mutation, yet environmental influences are key in the pathogenesis of the disease.¹³⁶

Furthermore, the expense of genomic analysis or biomarker diagnosis is another constraint in terms of access.¹³⁷ To overcome these obstacles, analyses of data pooling across tens, hundreds, or thousands of studies or biospecimens would be needed,^{138–140} and large-scale collaborative projects led by those like the Global Parkinson's Genetics Program (GP2) also the Alzheimer's Disease Sequencing Project (ADSP) for instance are seeking to grow genetic databases to enhance equity in the utility of precision medicine.^{141,142} Newer advancements like AI and multi-omics integration will further amplify precision medicine applications.^{143,144} Algorithms powered by AI can analyze large datasets to discover new patterns of disease and anticipate how individuals will respond to treatment.^{145,146} On the other hand, multi-omics, particularly genomics, proteomics, and metabolomics, can offer a comprehensive understanding of diseases. Thus, it has the potential for discovering new therapeutic targets. In line with this latter observation, metabolomic profiling is being employed to characterize distinct metabolotypes in patients with amyotrophic lateral sclerosis, thereby providing the groundwork for personalized metabolic therapies.¹⁴⁷ Precision medicine represents a revolutionary model for research and treatment of neurodegenerative disorders.^{148,149} It combines the power of genomics, biomarkers, and AI techniques to provide for earlier diagnosis, more focused treatments, and better-tailored preventative strategies.^{123,150} Nevertheless, the challenges associated with tissue diversities, expenses, and availability have to be addressed for more extensive use. Progress is ongoing and continued through collaborative research and technological development, with promise for better and more personalized treatment for neuro-degenerative patients.

8. Impact of neuromarkers on neuroinflammation

Neuromarkers are quantifiable bio-indicators that offer information about neurological functioning, diseases, or treatment effects. Within the frame of neuroinflammation, markers like cytokines (tumor necrosis factor- α , interleukin-6), glial fibrillary acidic protein, and neurofilament light chain play a crucial role in the diagnosis and monitoring of brain inflammation.^{151–153} These biomarkers can be found in blood-CSF or detected using neuroimaging approaches, leading to early diagnosis and monitoring of diseases such as multiple sclerosis, Alzheimer's disease, and traumatic brain injury.^{154–156} Recent advances in proteomics and genomics have also facilitated the discovery of new neuromarkers, providing new potential targets for the intervention.^{157,158} Neuroinflammation is a multifaceted response of the immune system within the central nervous system that can exert protective and damaging effects.¹⁵⁹ Whilst acute inflammation is beneficial for repairing damaged tissue, chronic inflammation of the nervous system leads to Parkinson's and Alzheimer's diseases.^{160,161} At the center of this lies the participation of microglia and astrocytes that generate pro-inflammatory cytokines and release reactive oxygen species that propagate Neuronal damage.^{162,163} Neuroimaging techniques using PET scans with TSPO (translocator protein) ligands can image neuroinflammation activity, enabling non-invasive tracking of the progression of the disease.^{164,165} Such a piece of knowledge could be of

importance for designing anti-inflammatory therapies aiming at reducing neurodegeneration. Neuromarkers are an attractive focus for pharmacological intervention in neuroinflammation.^{166,167} For instance, monoclonal antibodies to pro-inflammatory cytokines (e.g., drugs against interleukin-6) are also being tested in neurodegenerative and autoimmune diseases.¹⁶⁸ Furthermore, biomarkers, such as S100B and tau proteins, are contributing toward routine stratification of patients and personalized medicine approaches.^{20,69} Research future will integrate multimodal biomarker panels with machine learning to improve diagnostic sensitivity and treatment efficacy.^{169,170} As neuromarkers and neuroinflammation become more clearly defined, these tools may soon revolutionize early interventions and personalized medicine in neurology.

9. Role of neuromarkers in Parkinson's disease

Neuromarkers are measurable biological markers that signify a path to understanding a neurological disorder, e.g., Parkinson's disease.¹⁷¹ Neuromarkers have been defined among Parkinson's disease patients as genetic, biochemical, and neuroimaging-based parameters that allow early diagnosis, assessment of disease progression, and evaluation of treatment response.^{172,173} For instance, alpha-synuclein aggregates in the CSF and blood, in association with dopamine transporter imaging, may be candidate neuromarkers. Modern neuroimaging modalities, such as fMRI and PET, allow specific brain patterns of Parkinson's disease to be identified, thus increasing diagnostic accuracy.^{174,175} Wearable sensors to monitor akinetic/rigid movements or tremors are also becoming a digital neuromarker to monitor disease progression, and provide real-time data.^{176,177}

Notwithstanding recent progress, validation and standardization of neuromarkers for Parkinson's disease is difficult.¹⁶³ Interpatient variability in biomarker expression and the absence of standardized diagnostic cutoffs inhibit their clinical application. Efforts continue to search for better neuromarkers that are less invasive, e.g., metabolomic profiles and microRNA signatures in biofluids.^{178,179} Moreover, AI that integrates multimodal genetic, imaging, and clinical data may improve prediction accuracy and individual personalized treatments.¹⁸⁰ With continued development and validation, confirmed neuromarkers could prove to be the key to the early intervention and treatment of Parkinson's disease (Fig. 3).

10. Impact of neuromarkers in Alzheimer's disease

Neuromarkers are biological measures that aid in the early detection, diagnosis, and monitoring of neurological diseases such as Alzheimer's disease.^{33,181} In Alzheimer's disease, the main neuromarkers are A β plaques, tau proteins, and neurofilament light chain, which can be measured in the CSF and by advanced neuroimaging methods such as PET.^{182,183} These objective metrics may help detect the progression of the disease at an early point before the patient becomes symptomatic. Recent efforts have further explored blood-based markers for the detection of Alzheimer's disease, such as p-tau181 and p-tau217, offering less invasive and perhaps more scalable diagnostic options.¹⁸⁴

The evidence related to plasma phosphorylated tau isoforms such as p-tau 217 and p-tau 231, which have high diagnostic accuracy for differentiating Alzheimer's disease from other neurodegenerative diseases and are gaining attention as blood-based biomarkers.^{185,186} Novel biomarkers related to neuroinflammation glial fibrillary acidic protein and synaptic dysfunction (i.e. neurogranin) were also incorporated since these are becoming increasingly relevant in clinical practice.^{187,188} In addition, α -synuclein seed amplification assays and TAR DNA binding protein-43 plasma assays are also described as encouraging assays to discriminate Parkinson's disease and Braintemporal lobar degeneration, respectively.^{189,190} These additions enrich the discussion by mirroring the continued move to multi-marker, blood-based precision diagnostics for neurodegenerative diseases.¹⁹¹ The discovery of

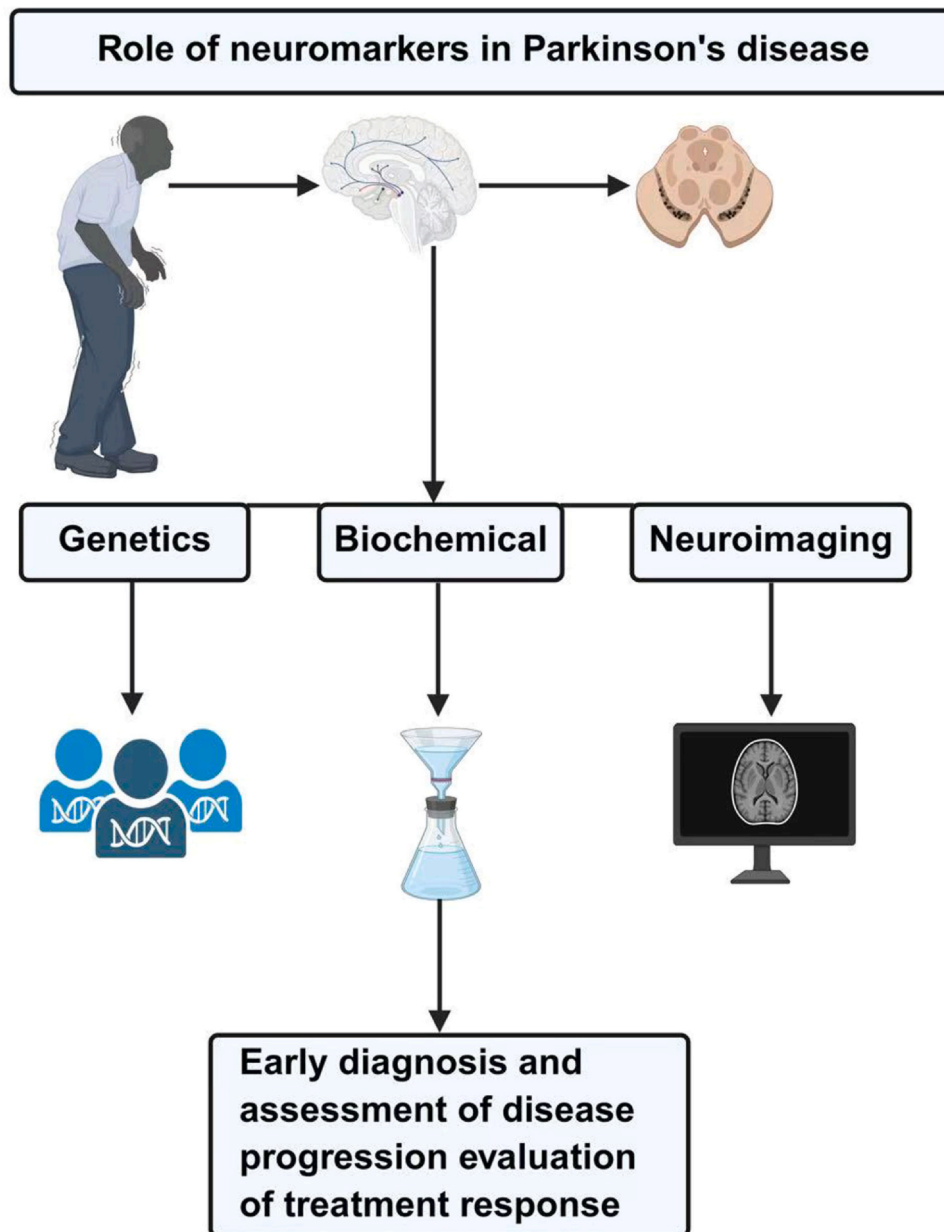


Fig. 3. Role of neuromarkers in Parkinson's disease. This figure illustrates the multidimensional framework through which neuromarkers contribute to the diagnosis and management of Parkinson's disease. The central representation of Parkinson's disease pathology shows neuronal degeneration within the substantia nigra and associated brain regions. Three major categories of neuromarkers are highlighted: genetic, biochemical, and neuroimaging. Created using BioRender.com.

sensitive and specific neuromarkers could be critical for advances in the diagnosis and monitoring of the therapeutic response of Alzheimer's disease as well as progression to personalized medicine.^{20,192}

11. Role of neuromarkers in autism disease

Newborn blood spots contain possible neuromarkers for autism spectrum disorder and carry the exciting potential for early intervention and individualized treatment.¹⁹³ The reports on event-related brain activity in autism spectrum disorder have identified certain neurophenotypes, such as altered gamma-band oscillations and decreased default mode network connectivity, which are linked with the social-cognitive dysfunction observed in autism spectrum disorder cases.^{194,195} fMRI has also identified abnormal activation in the fusiform gyrus and superior temporal sulcus, regions fundamental to social processing.¹⁹⁶ However, advances in machine learning and multimodal imaging combining EEG, fMRI, and genetic data hold potential for

improving diagnostic accuracy.^{197,198} These neuromarkers not only enhance our understanding of the neural correlates of autism spectrum disorder but also make the path for more objective, biomarker-based diagnostic procedures, which, in turn, help to diminish the dependence on behavioral evaluations.^{199,200}

12. Impact of neuromarkers on Huntington's disease

Huntington's disease is an autosomal dominant, progressive neurodegenerative disease characterized by CAG trinucleotide repeat expansion in the HTT gene and motor, cognitive, and psychiatric manifestations.^{201,202} Recent advances in neuroimaging and biofluid analysis have put forward candidate neuromarkers to track disease progression and predict the onset of symptoms.^{203,204} MRI imaging has identified early regional atrophy of the striatum and cerebral cortex and, with the use of diffusion tensor imaging, has shown evidence of white matter degeneration.²⁰⁵ In addition, biomarkers have been

Table 1

Different neurodegenerative diseases and their key neuromarkers.

Diseases	Neuromarkers	Ref.
Alzheimer's	Amyloid beta 42 (A β ₄₂) total tau protein in cerebrospinal fluid tau (CSF), Phosphorylated tau at threonine p-tau181, neurofilament light, Apolipoprotein E epsilon 4 genotype Apolipoprotein E ϵ 4 genotype.	220
Huntington's	Mutant Huntingtin protein mHTT protein, cytosine–adenine–guanine trinucleotide repeat expansion CAG, neurofilament light.	221
Parkinson's	Alpha-synuclein protein α -synuclein CSF (CSF/blood), Protein deglycase DJ-1, Leucine-Rich Repeat Kinase 2 gene mutations mutations, Glucocerebrosidase gene mutations GBA mutations.	222
Autism	Oxytocin, Mutations in the SH3 and multiple ankyrin repeat domains 3 gene mutations, Gamma-Aminobutyric Acid and Glutamate/glutamate imbalance, Interleukin-6 (inflammatory markers)	223

Table 2

Different neurodegenerative diseases and their key neuroimaging techniques and clinical utility.

Disorder	Neuroimaging techniques	Clinical utility	Ref.
Alzheimer's disease	Magnetic Resonance Imaging (MRI), Positron Emission Tomography (PET) (used for imaging Amyloid-beta (A β) and Tau protein deposits) PET (A β , Tau), Functional Magnetic Resonance Imaging (fMRI).	Early diagnosis, subtype differentiation, and treatment monitoring.	151,155,224
Parkinson's disease	MRI, PET (dopamine tracers), fMRI.	Early diagnosis, subtype classification, and drug response prediction.	128,151
Autism spectrum disorder	fMRI, Electroencephalography.	Identification of neural network dysfunction, treatment monitoring.	109,110,159
Huntington's disease	MRI, Diffusion Tensor Imaging, fMRI.	Tracking disease progression, functional network analysis.	106,159

identified in both the CSF and blood, such as the neurofilament light chain and mutant huntingtin proteins, which have been associated with disease severity and are possibly useful as readouts for therapeutic trials.²⁰³

Fluorodeoxyglucose positron emission tomography (FDG-PET) is an advanced functional neuroimaging technique that measures cerebral glucose metabolism to provides a sensitive marker of neuronal activity and synaptic function.^{206,207} The radiotracer Simbol 18 F-fluorodeoxyglucose is taken up by active neurons in proportion to their metabolic demand, thus allowing the visualization of regional metabolic changes that often precede structural brain alterations.^{207,208} In the past two decades, FDG-PET has become crucial in the diagnosis and monitoring of a wide range of neurodegenerative diseases. For instance, FDG-PET findings in Alzheimer's disease reveal characteristic posterior cingulate, temporoparietal, and precuneus hypometabolism.^{209,210} Similarly, FDG-PET identify disease-specific metabolic network patterns in Parkinson's disease and other parkinsonian syndromes to enable better differential diagnosis and assessment of disease progression.^{211,212} Owing to the depiction of early brain changes, the subsequent utilization of FDG-PET has tremendously revolutionized neurodegenerative disorders into an ever-increasingly precision medicine.^{100,213} These neuromarkers may have potential clinical use for early diagnosis of Huntington's disease and monitoring of treatment effects. However, the challenge of translating neuromarkers into the clinic for Huntington's disease remains, in light of the variability of disease course, and a need for protocol standardization is evident.^{214–216} While neurofilament light chain is viewed as an excellent biomarker, its specificity for Huntington's disease over other neurodegenerative disorders should be confirmed.²¹⁷ Novel methods have also been developed to improve predictive power, utilizing emerging technologies, such as PET imaging of mutant huntingtin aggregates and machine learning-based models that incorporate multiple modalities of data.^{218,219}

13. Limitations

Before concluding, it is necessary to make three notes about limitations that the field faces. While there have been considerable advancements in neuromarkers and neuroimaging for precision medicine, currently, they are limited in their clinical and research use. To begin with, different paradigms and imaging standards across research centers and laboratories create a problem with replicability and

standardization of the obtained results. In addition, most of the published neuromarker works have small samples, with insufficient strength and generalizability, especially within racial and ethnic differences. But there are critical limitations to these approaches as well. Thirdly, even though neuroimaging is invaluable for identifying structural and functional problems, it may not capture molecular and cellular processes moving through the brain as they continually change, such as dynamic shifts in transient neuroinflammation. Fourth, advanced neuroimaging implants, such as PET and fMRI, are expensive and difficult to acquire; incorporating MMMs into them alongside AI remains less than accurate as well. Lastly, limited longitudinal experiments have prevented researchers from creating extensive conceptual models of how novel types of neuromarkers may adjust and the extent to which they will correctly predict neurodegenerative and neurodevelopmental conditions. All of these drawbacks must be addressed in order to implement a novel set of neuroimaging and neuromarker methods for precision therapy.

14. Conclusion

Neuromarkers and neuroimaging pipelines are ready to revolutionize the field of precision medicine in diseases of the brain, such as neurodegenerative or neurodevelopmental conditions, with their more precise diagnosis, earlier presence, and understanding of pathophysiology (Tables 1 and 2). In comparison to other reviews, this manuscript uniquely focuses on working from evidence across diseases, including neuroinflammation, systems-level connectivity, and molecular imaging markers. Imaging and machine learning-based biomarker research with the combination of multiple models has great potential for clinical translation in terms of personalized medicine and better prognosis prediction. While these approaches also have their shortfalls, better research in the future would permit a more targeted approach, ongoing surveillance with enhanced understanding of individual differences. Thus, this review has a unique perspective on the promise of neuromarkers for neurologic precision medicine as an ordinary clinical instrument.

15. Future perspective and challenges

The next frontier in neurology is the development of neuromarkers that facilitate preventive and accurate diagnosis of neurodegenerative

diseases and neurodevelopmental diseases such as Parkinson's disease, Alzheimer's disease, autism, and Huntington's disease. Novel neuroimaging modalities, such as high-resolution fMRI and PET images and AI based analytics, are converging toward personalized precision medicine. Nevertheless, we are facing challenges to harmonizing the use of biomarkers across populations with contrasted diversity at both clinical and epidemiological levels in terms of cost-efficiency in biomarker validation, as well as integrating multi-omics data for personalized treatments. What is more, the dynamic progression of diseases such as Alzheimer's disease, for which the underlying pathology starts many decades before the first clinical symptoms become apparent, highlights the necessity for longitudinal studies to confirm predictive neuromarkers. Solving these problems has the potential to transform early intervention and patient outcomes.

New research is finding that neuroinflammation is a key player in conditions such as Parkinson's and Alzheimer's diseases, paving the way for new treatments to quell inflammation. The challenge is to target neuroinflammation while preserving critical immune functions. For autism and Huntington's disease, the heterogeneity of presentation makes it harder to design biomarkers or treatments that work for everyone. Progresses in CRISPR, gene therapy, and autism research are bringing new hope for Huntington's disease, and they are looking at synaptic function and connectivity. A huge obstacle is getting lab discoveries into the clinic, and this requires multidisciplinary teamwork and strong clinical trials. Overcoming these hurdles will be central to realizing future diagnostics and treatments for brain disorders.

CRedit authorship contribution statement

Farnaz Norouziyan: Writing – original draft. **Nasrollah Moradikhor:** Writing – review & editing, Writing – original draft, Supervision, Investigation. **Adeel Ahmed Abbasi:** Writing – review & editing, Writing – original draft. **Farhat Jokhio:** Writing – original draft.

Data availability

Not applicable.

Declaration of Competing Interest

The authors declare no competing interests in this work.

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